Word2vec Gensim python library

cbow and skipgram

# !/bin/bash

kaggle datasets download datatattle/covid-19-nlp-text-classification

```
import os
from google.colab import userdata

os.environ["KAGGLE_USERNAME"] = userdata.get('KAGGLE_USERNAME')
os.environ["KAGGLE_KEY"] = userdata.get('KAGGLE_KEY')
```

```
!kaggle datasets download datatattle/covid-19-nlp-text-classification
```

```
Dataset URL: https://www.kaggle.com/datasets/datatattle/covid-19-nlp-text-classification
License(s): copyright-authors
Downloading covid-19-nlp-text-classification.zip to /content
100% 4.38M/4.38M [00:00<00:00, 73.7MB/s]
```

```
!unzip -q covid-19-nlp-text-classification.zip -d content
```

```
os.listdir("./content")
```

```
['Corona_NLP_test.csv', 'Corona_NLP_train.csv']
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline

import torch
import torch.nn as  nn
```

```
dataset = pd.read_csv(r'./content/Corona_NLP_train.csv', encoding='latin1')
```

```
dataset.head()
```

| | UserName | ScreenName | Location | TweetAt | OriginalTweet | Sentiment |
|---|---|---|---|---|---|---|
| **0** | 3799 | 48751 | London | 16-03-2020 | @MeNyrbie @Phil_Gahan @Chrisitv https://t.co/i... | Neutral |
| **1** | 3800 | 48752 | UK | 16-03-2020 | advice Talk to your neighbours family to excha... | Positive |
| **2** | 3801 | 48753 | Vagabonds | 16-03-2020 | Coronavirus Australia: Woolworths to give elde... | Positive |
| **3** | 3802 | 48754 | NaN | 16-03-2020 | My food stock is not the only one which is emp... | Positive |
| **4** | 3803 | 48755 | NaN | 16-03-2020 | Me, ready to go at supermarket during the #COV... | Extremely Negative |

```
import re
texts=[]
for i in range(0,len(dataset)):
  text = re.sub('[^a-zA-Z]' , ' ', dataset['OriginalTweet'][i])
  text = text.lower()
  text = text.split()
  x = len(text) if text.count('https') ==0  else text.index('https')
  text = text[: x ]
  text = [t for t in text if not t=='https']
  text = ' '.join(text)
  texts.append(text)
```

```
texts[0]
```

Show hidden output

```
!pip install gensim
```

```
Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8.4 kB)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.5.1)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim) (2.1.2)
Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 27.9/27.9 MB 63.8 MB/s eta 0:00:00
Installing collected packages: gensim
Successfully installed gensim-4.4.0
```

```
from gensim.models import Word2Vec
```

```
sentences = [line.split() for line in texts]
w2v = Word2Vec(sentences, vector_size=100, window=5, workers=4, epochs=10, min_count=5)
```

```
print(sentences[:5])
```

```
[['menyrbie', 'phil', 'gahan', 'chrisitv'], ['advice', 'talk', 'to', 'your', 'neighbours', 'family', 'to', 'exchange', 'phone', 'numbers', 'create', 'contact', 'list', 'with', 'phone', 'numb
```

```
words = list(w2v.wv.index_to_key)
```

```
print(words)
```

```
['the', 'to', 'and', 'of', 'covid', 'a', 'in', 'coronavirus', 'for', 'is', 'are', 'i', 'you', 'on', 's', 'this', 'prices', 'at', 'it', 'food', 'supermarket', 'we', 'store', 'that', 'with', '
```

Now say word embeddings for "computer"

```
print(w2v.wv['computer'])
```

```
[ 0.17623939 -0.0649346   0.02180911 -0.08314048 -0.21801828 -0.26014817
 -0.17201686  0.26603958  0.12624052  0.18746084 -0.09201432  0.03048269
  0.02571641  0.26074713  0.14454576 -0.01562333  0.22457618 -0.28245917
 -0.07130527 -0.47806144  0.64517933 -0.10774837  0.06236184 -0.29114887
  0.01921001  0.05273705 -0.4576319  -0.01497949 -0.01280056 -0.04099673
  0.22447446  0.282011   -0.09085299 -0.10213461 -0.01499699 -0.04299729
 -0.30544642 -0.05537534 -0.16529456 -0.2907513  -0.15991525 -0.2791086
  0.01638102  0.10667728  0.31720597 -0.03479983 -0.04840082  0.11923193
  0.03896172  0.10486746  0.04874977 -0.06519393 -0.17564623 -0.15756376
 -0.14551121 -0.18123373  0.04590392  0.13435027  0.08161922 -0.02646526
  0.00631981 -0.06050677 -0.12641366  0.11775038 -0.3669262   0.3897384
 -0.05435329  0.21953873  0.15250401  0.22841038 -0.02430989 -0.26397344
  0.15792704 -0.11663277  0.06251925  0.01354926 -0.05250515  0.0660618
 -0.04096726  0.05108177  0.00146483 -0.3015041  -0.2922049   0.16899134
  0.04524283 -0.12487379 -0.04423115  0.13914403  0.14295031  0.03563055
 -0.00718339  0.13331479 -0.04834007 -0.09489561  0.09688634  0.16484258
  0.14520009 -0.02954005 -0.25808054 -0.13850309]
```

Now to find similarity between words

```
w2v.wv.similarity('biden','corona')
```

```
np.float32(0.268311)
```

```
w2v.wv.similarity('death','corona')
```

```
np.float32(0.19740698)
```

Now finding similar words to our input

```
print(w2v.wv.most_similar('modi'))
```

```
[('imran', 0.830615758895874), ('narendra', 0.8089490532875061), ('ji', 0.7938373684883118), ('honorable', 0.7687804102897644), ('narendramodi', 0.7565923929214478), ('nsitharaman', 0.755835
```

```
print(w2v.wv.most_similar('trump'))
```

```
[('putin', 0.780602216720581), ('donald', 0.7702961564064026), ('president', 0.7601224780082703), ('realdonaldtrump', 0.7549073696136475), ('gop', 0.7453668117523193), ('potus', 0.7194048762
```

```
print(w2v.wv.most_similar('corona'))
```

```
[('stayhome', 0.7419095635414124), ('flattenthecurve', 0.7253080606460571), ('coronaviruspandemic', 0.7157506346702576), ('washyourhands', 0.709255039691925), ('lockdowneffect', 0.7054296135
```

Now to find relationships

x:y then z:?

```
print(w2v.wv.most_similar(positive=['russian', 'russia'], negative=['arab']))
```

```
[('saudi', 0.7413305640220642), ('arabia', 0.7271626591682434), ('putin', 0.716482937335968), ('trump', 0.6990422010421753), ('opec', 0.6472818851470947), ('saudis', 0.6047244668006897), ('m
```

```
print(w2v.wv.most_similar(positive=['narendramodi', 'indian'], negative=['rahulgandhi']))
```

```
[('govt', 0.5740652680397034), ('hul', 0.5732787847518921), ('india', 0.5683450698852539), ('delayed', 0.5553766489028931), ('congress', 0.5528779625892639), ('pmoindia', 0.5479452610015869)
```

Finding the odd one out

```
w2v.wv.doesnt_match(['modi','rahulgandhi','putin'])
```

```
'putin'
```

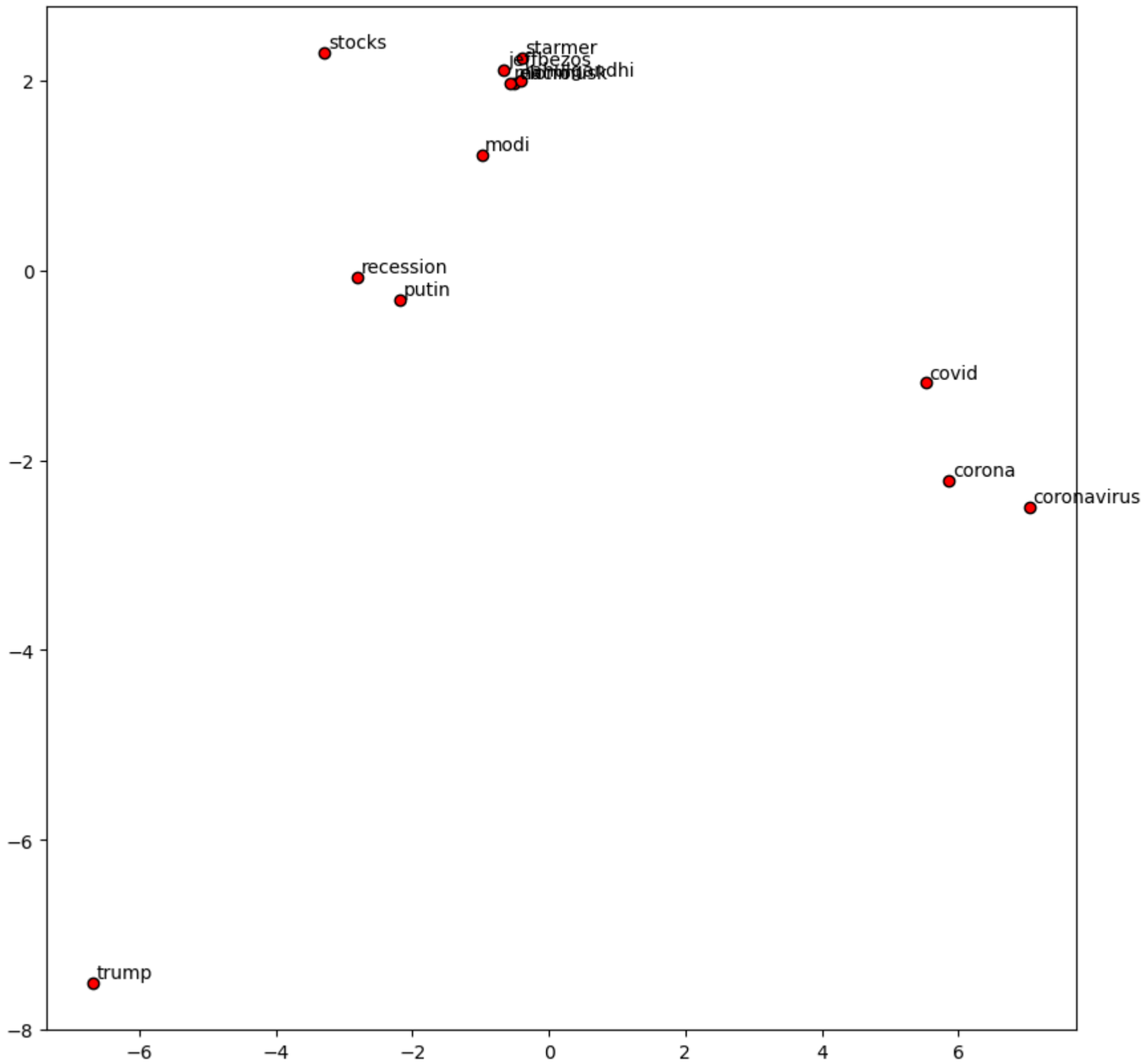
```
w2v.wv.doesnt_match(['corona','bjp','modi','india'])
```

```
'corona'
```

Visualization

```
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE
import matplotlib.pyplot as plt

def display_pca_scatterplot(model, words=None, sample=0):
  # Get all words from the model's vocabulary
  all_vocab_words = list(model.wv.index_to_key)

  if words is None:
    if sample > 0:
      words_to_plot = np.random.choice(all_vocab_words, sample, replace=False)
    else:
      words_to_plot = all_vocab_words
  else:
    # Filter user-provided words to only include those present in the model's vocabulary
    words_to_plot = [w for w in words if w in model.wv.key_to_index]
    if not words_to_plot:
        print("None of the provided words are in the model's vocabulary.")
        return

  # Get the word vectors for the words to plot
  word_vectors = np.array([model.wv[word] for word in words_to_plot])

  # Perform PCA to reduce dimensions to 2
  twodim = PCA(n_components=2).fit_transform(word_vectors)

  plt.figure(figsize=(10, 10)) # Increased figure size for better readability
  plt.scatter(twodim[:, 0], twodim[:, 1], edgecolors='k', c='r') # Fixed twodim[:.0] to twodim[:,0]
  for word, (x, y) in zip(words_to_plot, twodim):
    plt.text(x + 0.05, y + 0.05, word)
  plt.show()
```

```
display_pca_scatterplot(w2v, ['jeffbezos','elonmusk','corona','coronavirus', 'covid', 'starmer','putin','modi','trump','rahulgandhi','recession','stocks','macron'])
```




save the model

```
import gensim.downloader
print(list(gensim.downloader.info()['models'].keys()))
```

```
['fasttext-wiki-news-subwords-300', 'conceptnet-numberbatch-17-06-300', 'word2vec-ruscorpora-300', 'word2vec-google-news-300', 'glove-wiki-gigaword-50', 'glove-wiki-gigaword-100', 'glove-wik
```

Start coding or generate with AI.

new model

```
google_news = gensim.downloader.load('word2vec-google-news-300')
```

```
[==================================================] 100.0% 1662.8/1662.8MB downloaded
```

```
google_news.most_similar('Twitter')
```

```
[('twitter', 0.89089035987854),
 ('Twitter.com', 0.7694422602653503),
 ('Facebook', 0.7665740251541138),
 ('tweet', 0.7635129690170288),
 ('tweets', 0.7501592636108398),
 ('Tweet', 0.7295743227005005),
 ('tweeting', 0.7256590723991394),
 ('Tweets', 0.7165543437004089),
 ('TwitterTwitter', 0.705342173576355),
 ('Tweeting', 0.6768180131912231)]
```

Start coding or generate with AI.

saving our corona virus model

```
w2v.save("word2vec-covid.model")
```

```
model = Word2Vec.load("word2vec-covid.model")
model.train([['hello', 'world']], total_examples=1, epochs=1)
```

```
WARNING:gensim.models.word2vec:Effective 'alpha' higher than previous training cycles
(2, 2)
```

END

Start coding or generate with AI.